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*Environ. Sci. Technol.*, **Just Accepted Manuscript** • DOI: 10.1021/acs.est.8b03511 • Publication Date (Web): 14 Jan 2019

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# Understanding enhanced microbial MeHg production in mining-contaminated paddy soils under sulfate amendment: Changes in Hg mobility or microbial methylators?

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30 Total word count: 6757

# ABSTRACT

Elevated methylmercury (MeHg) production in mining-contaminated paddy soils, despite of the high fraction of refractory HgS(s), has been frequently reported, while the underlying mechanisms are not fully understood. Here, we hypothesized that sulfate input, via fertilization, rainfall and irrigation, is critical in mobilizing refractory HgS(s) and thus enhancing Hg methylation in mining-contaminated paddy soils. To test this hypothesis, the effects of sulfate amendment on Hg methylation and MeHg bioaccumulation in mining-contaminated soil-rice systems were examined. The results indicated 28–61% higher net MeHg production in soils under sulfate amendment (50–1000 mg $kg^{-1}$), which in turn increased grain MeHg levels by 22–55%. The enhancement of Hg methylation by Hg mobilization in sulfate-amended soils was supported by two observations: (1) the increased Hg(aq) release from HgS(s), the dominant Hg species in the paddy soils, in the presence of sulfide produced following sulfate reduction and (2) the decreases of refractory HgS(s) in soils under sulfate amendment. By contrast, changes in the abundances/activities of potential microbial Hg methylators in different Hg-contaminated soils were not significant following sulfate amendment. Our results highlight the importance to consider enhanced Hg mobility and thus methylation in soils under sulfate amendment.


Key words: Methylmercury; Mercury; Bioaccumulation; Bioavailability; Rice

## Introduction

Mercury-contaminated soils in mine areas are usually characterized by low methylmercury (MeHg) concentrations despite high total Hg (THg) levels (e.g. MeHg/THg: 0.0009–0.03%).[1,2] This has been largely attributed to the predominance of refractory Hg species in mining-contaminated soils, especially HgS,[3] which is normally less available to microbial methylators.[4] However, the elevated MeHg/THg ratios reported in some mining-contaminated areas (up to 0.97% in soils and 1.5% in the estuary near the Idrija Hg mine, Slovenia,[5,6] and as high as 0.47% in rice paddy soils in Gouxi, Wanshan Hg mine area, China[7]) remain largely unexplained. Consequently, elevated soil MeHg levels (as high as 23 μg $kg^{-1}$ in Wanshan[8] and 80 μg $kg^{-1}$ in soils near the Podljubelj Hg mine, Slovenia[9]) enhance MeHg accumulation in crops, an observation that has raised concern regarding dietary exposure to MeHg. For instance, the massive accumulation of MeHg in crops in Wanshan mine area (as high as 174 μg $kg^{-1}$ in rice grains[10]) has been reported, although refractory HgS is dominant in Wanshan soils (64–94%).[11–13] These results emphasize the need for a better understanding of the factors that may lead to the mobilization of otherwise refractory Hg species (e.g., HgS(s)) in mining-contaminated soils and their subsequent methylation, especially considering the growing evidence of microbial methylation of some HgS compounds such as neutral HgS species (e.g., $HgS^0$, $Hg(SH)_2^0$)[14–16] and Hg-S nanoparticles (e.g., nano-HgS).[17]

This study examined the potential role of sulfate in mobilizing refractory Hg species and thus facilitating Hg methylation in mining-contaminated paddy soils. Sulfate amendment has been widely reported to enhance the activities of microbial methylators (especially sulfate reducing bacteria, SRB) in natural and agricultural wetlands,[18–20] with sulfate serving as an electron acceptor for SRB. Sulfate-impacted microbial abundances/activities were thus believed to be critical in controlling Hg methylation in

soils and sediments, and could be responsible for the enhanced Hg methylation under sulfate input.[21-26] However, whether sulfate and its reduction in flooded paddy soils play a role in Hg mobility is unknown. The potential changes in Hg mobility under sulfate amendment (e.g., via fertilization, rainfall or irrigation), in addition to microbial abundances/activities, may also affect microbial Hg methylation in mining-contaminated paddy soils. Theoretical calculations suggest that sulfide ($S^{2-}$), a main product of sulfate reduction, promotes the solubilization of Hg from refractory HgS minerals.[27] Sulfide-facilitated solubilization of HgS(s) has also been supported by experimental evidences.[14,16] For instance, THg and MeHg levels in dissolved pore water were highest in samples form the estuarine region near the Idriga mine, where S cycling conditions are ideal for HgS(s) mobilization.[28] These findings suggest that the input and subsequent reduction of sulfate could possibly mobilize HgS(s) in mining-contaminated paddy soils and enhance microbial Hg methylation, a hypothesis tested in the present work. The results would contribute to explaining the elevated soil MeHg levels measured in some Hg mining areas. This knowledge is especially relevant considering the continuous input of sulfate into paddy soils through sulfur fertilization and atmospheric deposition. For example, in Guizhou province, where the largest Hg mine in China is located, ~ 60 mg sulfate $kg^{-1}$ is annually introduced (via fertilization, rainfall or irrigation) into soils.[29]

Here, we examined the ability of sulfate input to mobilize refractory Hg(s) and thereby increase the production and bioaccumulation of MeHg in mining-contaminated soil-rice systems. Pot experiments, consisting of rice cultivation in mining-contaminated soils under sulfate amendment, were conducted to reveal potential changes in the MeHg concentrations of soils and rice plants in response to sulfate amendment. In addition, batch experiments, in which five different Hg-contaminated paddy soils containing various levels of sulfate were incubated in batch reactors, were carried out to further

investigate the mechanisms underlying the sulfate-amendment-induced changes in soil MeHg levels. Contrary to previous studies, in which microbial Hg methylation under sulfate amendment was mainly demonstrated by changes in SRB abundances/activities, we focused on both the changes in microbial methylators (indicated by SRB abundances/activities, and copy number of the *hgc*A methylation gene), and the potential changes in Hg/S speciation (quantified using synchrotron radiation techniques) and thus Hg mobility. The latter could be important considering that sulfate is reduced to elemental sulfur ($S^0$) or to $S^{2-}$ under flooded conditions, which may subsequently modify the speciation and mobility of Hg in soils, e.g., by complexation, precipitation, or adsorption.[30,31]

## Materials and methods

**Soil, chemicals and containers.** The soil used in the pot experiments was collected at a depth of 0–20 cm from a mining-contaminated paddy field in the Wanshan Hg mine area, Guizhou province, China (herein referred to as WS soil). The soil was air-dried, mixed, and sieved to an effective diameter of ≤ 2 mm. The THg and MeHg concentrations in WS soil were 80.8 ± 1.4 mg $kg^{-1}$ and 3.0 ± 0.5 μg $kg^{-1}$, respectively, and total sulfur and dissolved sulfate levels were 270 ± 10.1 and 140.9 ± 0.2 mg $kg^{-1}$. In batch experiments, five Hg-contaminated paddy soils containing various ambient sulfate levels were used: WS soil, Qingzhen (QZ) soil (THg: 55.6 ± 0.3 mg $kg^{-1}$; MeHg: 38.4 ± 0.13 μg $kg^{-1}$), Xunyang (XY) soil (THg: 32.9 ± 0.4 mg $kg^{-1}$; MeHg: 4.4 ± 0.07 μg $kg^{-1}$), Loudi (LD) soil (THg: 0.2 ± 0.0 mg $kg^{-1}$; MeHg: 1.1 ± 0.1 μg $kg^{-1}$), and Guiyang soil (THg: 0.2 ± 0.0 mg $kg^{-1}$; MeHg: 0.08 ± 0.03 μg $kg^{-1}$). QZ soil was collected from an industrially contaminated site in Guizhou province and contained high sulfate levels (2020.5 ± 10.8 mg $kg^{-1}$). XY soil was obtained from Xunyang (Shannxi province), a

major Hg mining area in China and had relatively lower sulfate levels (158.2 ± 0.7 mg $kg^{-1}$) than QZ soil. LD (in Hunan province) and GY (in Guizhou province) soils were collected from the control sites in provinces where major Hg mines are located. Other soil characteristics are listed in Table S1. Solid speciation of Hg in different soils (i.e., geochemical fractions of Hg, determined by sequential extraction, details described below) is listed in Table S2.

The chemicals used in this study are listed in Table S3. Their background THg levels were below the detection limit (<0.01 μg $kg^{-1}$). The polypropylene centrifuge tubes used in this study (Corning, USA), including for sample digestion/dilution and as batch reactors, were considered Hg-free, as the amount of THg in 2% $HNO_3$ after rinsing the tubes was below the detection limit (<0.01 μg $L^{-1}$).

**Pot experiments.** Pot experiments, in which rice was cultivated in WS soil, were carried out to explore the potential effects of sulfate amendment on soil MeHg levels and on MeHg accumulation in rice. Each of the five treatments, one without (CK) and four with (50, 100, 500, and 1000 mg S $kg^{-1}$, abbreviated as 50S, 100S, 500S and 1000S) sulfate amendment, consisted of three replicate pots (Table S4). The amended doses were within the range reported in Wanshan (e.g., 8–767 mg S $kg^{-1}$)[32] and other mine areas (e.g., 1600–37400 mg S $kg^{-1}$ in Mieres and Pola de Lena, Spain,[33] and 1100 mg S $kg^{-1}$ in Almadén, Spain[34]). At the same time, rice plants were cultivated in low-Hg soil (THg: 0.09 ± 0.01 mg $kg^{-1}$) to explore the potential contribution of the foliar uptake of atmospheric Hg on the Hg levels in plants. Details of rice cultivation could be found in Supplementary Information, text.

At harvest, the entire plant was collected, rinsed thoroughly with DI water, and divided into root, straw (stalk and leaf) and grain (brown rice). The tissues were then

152 air-dried, weighed, and their THg and MeHg contents analyzed as described below. In
153 addition, rhizosphere soils were sampled, placed into tubes, and freeze-dried. They were
154 then ground into fine powders, sieved through a 150-μm mesh, and stored at –20 °C until
155 they were used in analyses of Hg and S speciation by X-ray absorption near-edge
156 structure spectroscopy (XANES, details described below) and in the quantification of
157 their MeHg levels and Hg geochemical fractions (described below).
158
159 **Batch experiments.** Sulfate-amendment-induced changes in microbial Hg
160 methylation in soils were elucidated by quantifying Hg mobility (batch experiment I) and
161 the abundances/activities of potential microbial methylators (batch experiment II).
162 Considering that HgS(s) is the dominant Hg species in WS and many other
163 mining-contaminated paddy soils[11–13] and that sulfate reduction under reduced conditions
164 would result in high levels of sulfide ($S^{2-}$),[35] the potential interactions between HgS(s)
165 and $S^{2-}$ at different sulfide: Hg ratios warranted investigation. The analysis was expected
166 to shed light on the changes in Hg mobility and thus Hg methylation in
167 mining-contaminated soils under sulfate amendment. While HgS(s) dissolution in the
168 presence of $S^{2-}$ has been reported before,[14,16,27] we compared Hg release from HgS(s) and
169 spiked Hg(ads) in soils under $S^{2-}$ input. These would help explore the co-effects of Hg
170 speciation (e.g., HgS and other more mobile Hg species) and $S^{2-}$ on Hg mobility in soils.
171 Accordingly, in batch experiment I, Hg(aq) release from HgS(s) particles in the presence
172 of different levels of $S^{2-}$ was quantified. Specifically, 6 mg of HgS(s) particles were
173 added to each tube together with different doses of sodium sulfide ($Na_2S·9H_2O$, molar
174 ratios of sulfide: Hg = 0, 1:8, 1:4, 5:4, 5:2 and 5:1, dissolved in ultrapure water). The
175 tubes were sealed, incubated in darkness for 7 days at 28 °C, centrifuged at 5000×*g* for
176 20 min, and THg(aq) concentrations in the supernatant were then determined as described

177 below. In addition to the effect of $S^{2-}$ on the mobility of refractory HgS(s), the potential
178 influence of $S^{2-}$ on more mobile Hg species in an Hg-spiked soil was investigated.
179 Specifically, Hg(aq), as mercury chloride in ultrapure water, was added to low-Hg paddy
180 soil (THg: 2.4 ± 0.2 μg $kg^{-1}$, MeHg: 1.2 ± 0.2 μg $kg^{-1}$, sieved through a 100-μm mesh)
181 collected from Yixing, Jiangsu province, China (herein referred to as YX soil), to reach a
182 spiked concentration of 50 mg Hg $kg^{-1}$ soil, comparable to the amount in WS soil. After
183 equilibration of the soil for 12 hours at 25 °C, different amounts of sodium sulfide (molar
184 ratios of sulfide: Hg = 0, 1:8, 1:4, 5:4, 5:2, 5:1, 10:1 and 50:1) were added and the
185 mixtures were incubated for 18 hours at 25 °C. A much shorter incubation time (18 hours
186 vs. 7 days for HgS) was used in this experiment to avoid the potential methylation of the
187 spiked Hg(ads) in the YX soil during the incubation. At the end of the incubation period,
188 the THg(aq) concentrations in the supernatants were quantified.

189 Potential changes in microbial methylators under sulfate amendment were explored
190 in batch experiment II. Specifically, a series of microcosm experiments (mimicking
191 rhizosphere environments during rice cultivation)[36] was conducted with four treatments,
192 each set up in triplicate: CK (unamended WS soil), 500S (WS soil amended with sulfate:
193 500 mg S $kg^{-1}$), 1000S (WS soil amended with sulfate: 1000 mg S $kg^{-1}$), and CK-SRB
194 (WS soil to which sodium molybdate, a specific inhibitor of SRB, was added). These
195 batch experiments enabled the exploration of the earlier (days 0–25) and probably more
196 dynamic phases of microbial MeHg production in WS soil under sulfate amendment,
197 while precluding the need to intensively sample the soils during the early stage of rice
198 growth in the pot experiments. Briefly, WS soil (10 g) was placed in the tubes, to which
199 sulfate, sodium molybdate ($Na_2MoO_4$, final concentration of 20 mM), and 30 mL of
200 ultrapure water were also added. In all treatments, sodium lactate (final concentration of
201 5 mM) was provided as an electron donor, to stimulate microbial activity by mimicking

the labile carbon sources of the root exudates of rice.[36,37] The tubes were capped, sealed with Parafilm™, and incubated in darkness at 28 °C for 25 days, simulating the rhizosphere environment in rice paddy fields (see[36–38] for further information). During the incubation, all tubes were manually mixed twice per day. On days 0, 5, 10, 15, 20, and 25, three replicate tubes were opened, and the pH and Eh of the overlying water were measured in a glovebag (Atmos Bag, Sigma Aldrich) filled with nitrogen gas, using HQ30d multi-parameter meters (HACH, USA). Thereafter, the mixtures were centrifuged at 2000×*g* for 20 min, the supernatant was removed, and the soil was analyzed for MeHg (specified below), SRB abundances (on days 0 and 15 only) based on the most probable number method,[39] and *hgc*A gene copy number (indicative of potential microbial methylators, measured on day 15 only), according to Liu et al.[21] Further details on *hgc*A quantification was provided in Supporting Information, text. Besides, changes in dissolved sulfate levels (day 5, 10, or 15 vs. day 0; %), which may provide an estimate of sulfate reduction rates and SRB activities,[22,25] were calculated by quantifying dissolved sulfate levels (IC, Thermo ICS-2100; Thermo Fisher Scientific, NY) on day 0, 5, 10, and 15. Along with WS soil, the QZ and XY paddy soils were amended or not with sulfate (500 mg S $kg^{-1}$), and subjected to a 20-day incubation as described above. Concentrations of MeHg in soils were then quantified on days 0, 10 and 20. Copy numbers of *hgc*A methylation gene were determined on day 15 like WS soil. Besides the heavily contaminated WS, QZ and XY soils (HgS/Soil total Hg=64.7–75.8%), LD and GY paddy soils (HgS/Soil total Hg=23.4–28.5%) were amended or not with sulfate (500 mg S $kg^{-1}$) and incubated for 20 days like above, to explore the potentially different responses of microbial Hg methylation to sulfate input in soils with various HgS/Hg ratios. Concentrations of MeHg in LD and GY soils were then quantified on day 0 and 20.

**Analysis of Hg and S speciation in soils.** Hg and S speciation in WS soil, amended or not with sulfate (from pot experiments, at harvest), was analyzed by XANES according to Li et al.[40] and Xu et al.[41] Hg $L_3$-edge XANES experiments were performed on the BL14W beamline at the Shanghai Synchrotron Radiation Facility (SSRF). Due to limited available hours for the XANES, the analysis was carried out only on the CK and 500S treatments in pot experiments. The reference compounds were $HgCl_2$ (Hg-Cl-bonding form), $Hg(GS)_2$ (i.e., Hg-Glutathione, commonly used as reference compound in XANES analysis, to indicate S-Hg-S bonding species[42]), HgS(s), and MeHgCl ($CH_3$-Hg-Cl bonding form). Hg speciation in soils was also assessed by quantifying the geochemical fractionation of Hg based on the commonly used sequential chemical extraction method (SCE) of Jeyakumar et al.,[43] modified by Wang et al.[44] The distribution of Hg in various biogeochemically relevant fractions, i.e., soluble and exchangeable fraction (SEXF), specifically sorbed fraction (SPAF), iron/manganese oxide bound fraction (IMOF), organic matter bound fraction (ORGF), and residual fraction (RESP), was determined by sequential extraction followed by centrifugation at 5000×*g* for 30 min after each extraction step. Considering that SEXF and SPAF are commonly recognized as the most mobile fractions,[44] the two fractions were combined in Table S2.

Sulfur XANES of the soil samples and the reference materials was performed on the beamline 4B7A at the Beijing Synchrotron Radiation Facility (BSRF).[45] More details of XANES analysis of Hg/S speciation could be found in Supplementary Information, text.

**THg and MeHg analysis of rice tissues and soils.** THg levels in rice tissues (root, straw and grain) and soil samples were measured by inductively coupled plasma mass spectrometry (ICP-MS, Thermo Elemental X7, USA). The corresponding MeHg

measurements were made using cold vapor atomic fluorescence spectrometry (CVAFS, Brooks Rand, USA) following USEPA method 1630 (USEPA, 2001). Detailed information and data on the recoveries of THg and MeHg in the certified reference materials are provided in Supporting Information, text.

**Statistics.** Statistical analyses were conducted using SPSS, (version 16.0). Differences among treatments were assessed using Tukey's multiple comparison test of a one-way analysis of variance (ANOVA) ($p<0.05$).

## RESULTS

**Enhanced soil MeHg levels and MeHg accumulation in rice under sulfate amendment.** In the pot experiments, MeHg concentrations were elevated in soils under sulfate amendment (28–61%, vs. CK, Figure 1A), with a trend towards increasing soil MeHg levels with increasing amended sulfate doses. Similarly, MeHg concentrations in rice grains (Figure 1B) and whole plants (Figure S1C) were significantly higher ($p<0.05$) in treatments amended with higher sulfate levels (500S and 1000S) than in the CK control. In contrast, sulfate amended at various concentrations (0–1000 mg S $kg^{-1}$) had insignificant effects on plant biomass ($p>0.05$) (see Figure S2). Furthermore, the sulfate-induced changes in grain MeHg levels [$\Delta MeHg_{Grain}$, defined as: (grain MeHg level in sulfate-amended treatments − grain MeHg level in CK) *100% / grain MeHg level in CK] and the changes in soil MeHg levels [$\Delta MeHg_{soil}$, defined as: soil MeHg level in sulfate-amended treatments − soil MeHg level in CK) *100% /soil MeHg level in CK] increased linearly with the added sulfate doses (Figure 1C). The regression lines for

275 $\Delta MeHg_{Grain}$-added sulfate doses paralleled those of $\Delta MeHg_{soil}$-added sulfate doses
276 (Figure 1C). The correlation between $\Delta MeHg_{Grain}$ and $\Delta MeHg_{soil}$ was also significant
277 ($p<0.05$; Figure 1D) and followed a nearly 1:1 relationship, suggesting that increases in
278 grain MeHg concentrations were mainly due to increased MeHg levels in soils under
279 sulfate amendment.
280
281 **Increased Hg mobility in soils under sulfate amendment.** Similar to MeHg, the
282 inorganic Hg concentrations in grains (30–52% higher, sulfate amended vs. CK, Figure
283 S3A) and whole plants (25–50% higher, Figure S3D) were higher under sulfate
284 amendment, suggesting the increased mobility and bioavailability of inorganic Hg(ads) in
285 soils. Inorganic Hg levels in grains of rice plants cultivated in low-Hg soil (10 μg $kg^{-1}$)
286 were 14.1–21.5% of those cultivated in WS soil (46.6–70.9 μg $kg^{-1}$), demonstrating that
287 accumulated inorganic Hg in CK and sulfate amended treatments should be mainly from
288 soils instead of the atmosphere.
289 Meanwhile, XANES analysis of the soils in the pot experiments (Hg $L_3$-edge
290 XANES spectra; Figure S4) evidenced reduction in the refractory HgS(s) under sulfate
291 amendment (Table S5): The percentage of Hg as HgS species decreased to 45.6% in
292 500S (compared to 73.4% in CK) under sulfate input, while the percentage of
293 $Hg(GS)_2$-like (i.e., S-Hg-S) species increased to 37.1% in 500S (25.6% in CK). SCE data
294 also indicated that the addition of sulfate led to a decrease in the proportion of Hg in the
295 residual fraction (RESF, e.g., HgS[11–13]): The amount of Hg in RESF decreased from

296 64.7% in the CK control to 45.7% and 40.6% following amendments of 500 and 1000 mg
297 S $kg^{-1}$, respectively (Figure S5).

298 Besides pot experiments, evidence of increases in Hg mobility in soils following
299 sulfate reduction (e.g., to sulfides) was also obtained in batch experiment I. In the
300 absence of sulfide addition, the amount of dissolved Hg(aq) released from HgS(s) (the
301 principle Hg species in WS soil and many other mining-contaminated soils)[11–13] was
302 extremely low (0.04 mg $L^{-1}$, Figure 2A) but it increased sharply with increasing sulfide
303 levels: from 1.7 to 146.3 mg $L^{-1}$ when the molar ratio of sulfide: Hg increased from 1:8 to
304 5:1. Compared to refractory HgS(s), the release of newly loaded Hg(ads) from the YX
305 soil (Figure 2B) was less affected by the higher molar ratio of sulfide: Hg (from 1:8 to
306 5:1). However, sharp increases in Hg release were measured at higher sulfide: Hg ratios
307 (from 5:1 to 50:1).

308 The XANES results for sulfur in soils of the pot experiments may help explain the
309 sulfate-induced changes in Hg speciation and mobility, considering the interactions
310 between Hg and reduced sulfur species (e.g., $S^{2-}$ and FeS). The majority of sulfur in the
311 CK control was in the form of sulfate (66.2%) and sulfide (33.8%). However, in the
312 sulfate-amended paddy soils, sulfur as sulfate decreased to 43.9% while sulfur as sulfide
313 increased to 39.5% (Table S6 and Figure S6), confirming the dominance of sulfide
314 following sulfate reduction. Increases in FeS (18.1% vs. 0% in CK) in the
315 sulfate-amended soils were also determined.

316

317 **Minor effects of sulfate amendment on the abundances/activities of potential**
318 **microbial methylators.** Besides Hg speciation and mobility in soils, microbial
319 methylators are commonly believed to play a key role in Hg methylation,[20,46] a potential
320 investigated in batch experiment II. Consistent with the results of the pot experiments,
321 MeHg levels in WS soil (indicative of the net production of MeHg in soils) were
322 generally higher in the sulfate-amended soils than in the CK control, averaging 34%
323 higher in the 500S treatment and 35% higher in the 1000S treatment (Figure 3A). MeHg
324 levels in WS soil to which sodium molybdate had been added (CK-SRB) to inhibit SRB
325 activities were 42–61% lower than those in CK, evidencing the key role of SRB in
326 controlling net MeHg production in WS soil. However, the results of the microbial
327 analysis suggested that changes in microbial methylators (including SRB) may not be
328 primarily responsible for the increased soil MeHg levels under sulfate amendment: The
329 sulfate-amendment-induced increases in soil MeHg levels were highest on day 15
330 (64–78% higher vs. CK; Figure 3A) whereas the abundances of potential microbial
331 methylators in the soils were less affected by sulfate input. Specifically, the changes in
332 SRB abundances and *hgc*A copy numbers were insignificant ($p$>0.05, sulfate-amended vs.
333 CK, Figure S7A and S7B). Besides, changes in dissolved sulfate levels (%, compared to
334 day 0), indicating sulfate reduction and SRB activities, were not significantly affected by
335 sulfate amendment in the latter period of incubation (500S or 1000S vs. CK, $p$>0.05 on
336 day10 and 15, Figure S8B), further demonstrating the minor role of sulfate addition on
337 microbial methylators in WS soil. Sulfate addition also had minor effects on the Eh and

pH of the overlying water (Figure S9).

Similar to the WS soil, the MeHg levels in XY soils were significantly higher under sulfate amendment (500S vs. CK, $p<0.05$, Figure 3B). By contrast, sulfate amendment had insignificant effects ($p>0.05$) on both the MeHg concentration of the QZ soil, with elevated ambient sulfate levels (Figure 3C), and the *hgc*A copy numbers in either the XY or QZ soils ($p>0.05$, Figure S7B). Different from the mining-contaminated WS and XY soils with HgS as the dominant Hg species (HgS/Soil total Hg=64.7–75.8%), sulfate input could not enhance soil MeHg levels in LD and GY soils (HgS/Soil total Hg=23.4–28.5%, Figure S10 and Table S2). Especially, soil MeHg levels decreased in GY soil under sulfate amendment (500S versus CK, $p<0.05$, on day 20, Figure S10).

## DISCUSSION

The results of both the pot and the batch experiments clearly demonstrated that sulfate amendment could enhance net MeHg production in mining-contaminated paddy soils (WS soil: 28–61% and 43–67% higher than CK in pot and batch experiments, respectively; Figure 1A and Figure3A). The elevations in MeHg (54.1–74.8 μg $kg^{-1}$) and THg (100.6–145.8 μg $kg^{-1}$) levels that occurred in rice cultivated in sulfate-amended soils raise concerns about the impact on ecological and food safety. Hg levels in rice grains were higher than the national limit for Hg in grains (THg: 20 μg $kg^{-1}$) and within the ranges recently reported in Hg-contaminated areas in China (THg: 10.3–1120 μg $kg^{-1}$), [47] highlighting the need to better understand factors controlling Hg accumulation in

rice. The enhanced bioaccumulation of MeHg could be mainly attributed to the elevated MeHg levels in soil under sulfate amendment, in view of the close-to 1:1 relationship (Figure 1D) between the sulfate-induced increases in grain MeHg ($\Delta MeHg_{Grain}$) levels and soil MeHg ($\Delta MeHg_{soil}$). To our knowledge, this is the first report of elevated MeHg production in mining-contaminated paddy soils and thus enhanced MeHg accumulation in rice under sulfate amendment.

Unlike previous studies, which mainly focused on sulfate-induced increases in SRB abundances/activities, our study demonstrated that the mobilization of Hg in mining-contaminated WS soil facilitated the microbial transformation of Hg to MeHg and may thus be primarily responsible for increasing MeHg levels in soils under sulfate amendment.

**Sulfate amendment mobilizes Hg in soils and thus facilitates Hg methylation.** The mobilization of Hg in WS soils under sulfate amendment was evidenced by the results of pot experiments, together with XANES and SCE data: (1) While we did not directly quantify the changes in bioavailable Hg in soils under sulfate amendment, the increases in inorganic Hg accumulation in rice plants under sulfate amendment (Figure S3) provided evidence of increases in Hg mobility and bioavailability. (2) XANES analysis (Table S5 and Figure S4) and SCE data (Figure S5) revealed the reduction of refractory Hg(s) in WS soils under sulfate amendment. The mobilization of refractory HgS(s) in soils may subsequently enhance microbial transformation of soil Hg to MeHg, as determined in both the pot and the batch experiments.

While the sulfate-amendment-induced mobilization of refractory Hg in mining-contaminated paddy soils has not been reported before, the underlying mechanisms remain far from clear. According to the results of the XANES analysis, sulfate amendment led to elevated percentages of reduced sulfur species (e.g., sulfide and FeS, Table S6), previously shown to be the main species of sulfur in wetland sediments and paddy soils.[48] Sulfide could in turn facilitate Hg(aq) release from solid-phase HgS(s) (Figure 2A), the main Hg species in WS and many other mining-contaminated soils,[11–13] according to batch experiment I. The observations (i.e., increased Hg(aq) release from HgS(s), in the presence of sulfide) are consistent with the chemical reactions that occur in the dissolved phase (e.g., $Na_2S$ and dissolved HgS)[49] and with theoretical calculations: A thermodynamic model (Solgaswater (WinSGW) software, parameters listed in Table S7) was used to predict Hg(aq) dissolution from soil HgS(s) in the presence of different levels of $S^{2-}$. Results of the modelling indicate the increase of total dissolved Hg(aq) species following the increase of sulfide concentrations (Figure S11), in line with results obtained from batch experiment I.

The mobilization of Hg could be attributed to the formation of Hg-S complexes. (1) Previous reports showed that Hg and $S^{2-}$ could form a series of dissolved Hg-S complexes (e.g., $HgHS_2^-$, $Hg(HS)_2^0$, $HgS_2^{2-}$, HgS(0)[15]) and that HgS nanoparticles could also be generated during sulfate reduction.[17] Besides, majority of dissolved Hg was complexed by polysulfide, as predicted by the model (Figure S11), further indicating the formation of dissolved Hg-polysulfide species(aq) in the presence of $S^{2-}$, due to the

401 dissolution of HgS(s). (2) In addition, $S^0$, another major product of sulfate reduction,[48]
402 has been shown to facilitate the mobilization of Hg from HgS(s), by forming complexes
403 between Hg and polysulfides.[30,50] The availability of many of those Hg-S complexes (e.g.,
404 neutral HgS(0) and some Hg-polysulfide complexes) as well as HgS nanoparticles[51-54] to
405 microbial methylators was recently demonstrated. Our results, together with those
406 previous findings, therefore indicate that the amendment and subsequent reduction of
407 sulfate in flooded paddy soils facilitate Hg solubilization and the mobilization of HgS(s)
408 in WS soil, which in turn enhances Hg methylation by increasing the mobility and
409 availability of Hg to microbial methylators.

410 Besides the complexation between Hg and inorganic sulfur species (e.g., $S^{2-}$ and $S^0$),
411 soil- or plant-derived organic matter (OM) in paddy soils and thus the formation of
412 Hg-S-OM complexes may also play a role in Hg speciation and thus facilitate Hg
413 methylation in sulfate-amended soil,[17,55-58] considering that Hg-S-OM complexes may be
414 potentially available to microbial methylators.[57] However, more studies are needed to
415 explore the relative importance of inorganic sulfur and organic sulfur (e.g., OM-S), as
416 well as their co-effects, in affecting Hg speciation and methylation in
417 mining-contaminated soils. Further discussions could be found in Supplementary
418 Information, text.

419 While Hg was mobilized in the sulfate-amended mining-contaminated paddy soil
420 with a high HgS content, this may not be the case in other types of Hg-contaminated soils
421 in which Hg is more mobile (i.e., in YX, LD and GY soils). Unlike the HgS-dominated

422 WS soil, the solubilization of spiked Hg(ads) from the YX soil was less responsive to
423 sulfate amendment. When the sulfide:Hg ratio increased from 0 to 5:1, dissolved Hg(aq)
424 in the YX soil increased by a maximum of 1.6-fold whereas Hg(aq) release from HgS(s)
425 increased by up to 3656-fold (Figure 2A and Figure 2B). Interestingly, increases in the
426 sulfide:Hg ratios from 0 to 1:8 and from 5:2 to 5:1 decreased the amount of dissolved Hg
427 by 14% and 31% for YX soil, respectively (Figure 2B). This result can be explained by
428 the simultaneous and counteracting effects of: (1) the binding of soil-associated Hg(ads)
429 with $S^{2-}$ to form HgS(s) precipitates,[59] which would have reduced the mobility of Hg(ads)
430 in the soil, and (2) the sulfide-induced solubilization of Hg from HgS(s) (e.g., via the
431 formation of Hg-polysulfide complexes). Similarly, sulfate input could not enhance
432 microbial Hg methylation in LD and GY soils, in which Hg as HgS(s) was relatively low
433 (23.4–28.5%), and even reduced soil MeHg levels in GY soil (Figure S10). The decreases
434 in soil MeHg levels for GY soil under sulfate amendment could also be attributed to the
435 binding of mobile Hg(ads) with $S^{2-}$ following reduction of amended sulfate: Results of
436 sequential chemical extraction indicate that Hg in the residual fraction (e.g., HgS) in GY
437 soil increased from 28.5% in CK to 45.6% in 500S (Figure S12). In a previous study,
438 amendment with sulfate alone or sulfate with iron amendment was also shown to reduce
439 Hg mobility in Hg-spiked soils[59] and in hydroponic systems,[40] probably due to the
440 formation of refractory Hg species under sulfate amendment. Differently, the sharp
441 increases in dissolved Hg(aq) at higher sulfide:Hg ratios (5:1, 10:1 and 50:1) for
442 Hg-spiked YX soil presumably reflected the sulfide-induced mobilization of Hg from

443 HgS(s), considering at elevated sulfide:Hg ratios a majority of the mobile Hg in YX soil
444 may already be bound to amended $S^{2-}$. The observed differences in Hg mobility in the
445 refractory and mobile Hg pools following sulfate amendment of the soil indicated
446 differential effects of added sulfate on the mobility of Hg in mining-contaminated and
447 other types of soils in which Hg speciation (e.g., the percentage Hg as HgS) may be
448 substantially different.

449 **Minor changes in microbial methylators under sulfate amendment.** While
450 sulfate-induced increases in the abundances/activities of SRB are commonly thought to
451 drive the enhanced microbial MeHg production that occurs under sulfate amendment in
452 wetland sediments,[18–20] our results may suggest a lesser role for microbial methylators in
453 the studied soil: SRB numbers and copy numbers of the *hgc*A methylation gene in the
454 WS soil (indicative of the abundances of potential microbial methylators), as well as the
455 activities of SRB (indicated by changes in sulfate levels with time), were less affected by
456 sulfate amendment, despite the 1.6- to 1.8-fold increases in soil MeHg levels (Figure 3A
457 and Figure S7). Here, changes in dissolved sulfate levels with time were determined in
458 different treatments to provide an estimate of the activities of potential microbial
459 methylators under sulfate amendment, mainly because: (1) SRB was found to be the
460 principle microbial methylator in the tested WS soil (Fig.3A), and (2) Sulfate reduction
461 rates have commonly been reported to indicate SRB activities.[25] These results differ from
462 those of other studies in wetlands.[18–20] The inconsistency can be explained by the
463 relatively high ambient sulfate levels in the WS soil used in this study, in which

464 dissolvable sulfate concentrations were 140.9 ± 0.2 mg $kg^{-1}$. According to a previous
465 report, sulfate concentrations > 110 mg $kg^{-1}$ should not limit the growth of SRB.[60]
466 Therefore, the further addition of sulfate into the WS soil is unlikely to have significantly
467 enhanced the growth of SRB (the key microbial methylator in WS soil) and thus
468 microbial MeHg production. The inability of an excess supply of sulfate to facilitate
469 microbial MeHg production by SRB in a wetland soil at Yolo Bypass, California (USA)
470 was similarly reported.[61,62] Furthermore, insignificant relationship ($p$>0.05) was found
471 between soil MeHg levels and dissolved sulfate levels (Figure S13A-C) or between
472 changes in soil MeHg levels and changes in dissolved sulfate levels (day 5, 10 or 15 vs.
473 day 0, Figure S13D). These results may provide additional evidence that activities of
474 potential microbial methylators (e.g., SRB) may play a less important role in controlling
475 MeHg production under sulfate-adequate condition. While the limited data and short
476 incubation period (15 days) used in this experiment do not completely exclude an
477 enhancement of the abundances/activities of microbial methylators (especially SRB)
478 under sulfate amendment, our results nonetheless suggest that the changes in Hg
479 speciation and mobility induced by sulfate amendment could play an important role in net
480 MeHg production, particularly within the short-term following sulfate amendment.

481 Similar to WS soil, sulfate amendment of the XY soil, in which HgS(s) is also the
482 dominant Hg species, led to an increase in net MeHg production (75.8 ± 3.2%, Table S2).
483 Copy numbers of the *hgc*A methylation gene in XY soil were less affected by sulfate
484 amendment (Figure S7B). These results further evidence that, under sulfate-sufficient

conditions (158.2 ± 0.7 mg $kg^{-1}$ in XY soil, >110 mg $kg^{-1}$,[60]), changes in microbial methylators play a less important role in facilitating microbial MeHg production. However, the effects of sulfate amendment on soil MeHg levels in QZ soil were insignificant (Figure 3C), even though HgS was the main Hg species in this soil as well (67.26 ± 1.72%, Table S2). The difference can be explained by the high ambient sulfate levels in QZ soil (2020.5 ± 10.8 mg $kg^{-1}$), such that moderate increases in the amount of sulfate (500 mg $kg^{-1}$) may have only minor effects on either potential microbial methylators (indicated by *hgc*A copy numbers, Figure S7B) or Hg mobility and availability. Combining the results of the WS, YX and QZ soils revealed a positive linear relationship between the MeHg/THg ratios (indicative of the methylation potential) and the sulfate concentrations in soils ($R^2$ = 0.73, $p$<0.01, Figure S14 A), whereas the relationship between *hgc*A methylation gene copy number (indicating potential microbial methylators) and either MeHg/THg ratios or sulfate levels was not significant ($p$>0.05, Figure S14B and Figure 14C, respectively). Thus, under sulfate-sufficient conditions, sulfate-induced increases in microbial MeHg production cannot be mainly attributed to changes in the abundances of microbial methylators (e.g., SRB).

To conclude, our results raise concern that further inputs of sulfate into wetland systems with high inherent sulfate levels, as is the case in many Hg-mining contaminated areas, such as Wanshan, China,[32] Mieres and Pola de Lena (Spain),[33] and Almadén Hg mining district (Spain),[34] may mobilize refractory Hg species in soils, thus enhancing MeHg production and the risk of bioaccumulation. For instance in Wanshan, Guizhou

province, where the largest Hg mine in China is located, the annual input of sulfate (60 mg $kg^{-1}$)[29] is large enough to increase soil and grain MeHg levels by up to 61% and 44%, respectively, based on the linear relationship between the changes in the MeHg concentrations in soils and grains and the amended sulfate doses (Figure 1C, a 95% upper confidence limit was used to calculate the range of potential changes). Therefore, sulfur fertilization-facilitated MeHg production should be considered when assessing risk of MeHg bioaccumulation in Hg mining areas.

## ACKNOWLEDGEMENTS

Financial support was provided to YXG, ZYZ and JTZ by the National Natural Science Foundation of China (U1432241, 21377129, 21777162). HZ was supported by the National Natural Science Foundation of China (41673075). We thank the staff of BL 4B7A, 1W1B (BSRF) and BL14W, BL15U (SSRF) for their assistance. Finally, we are very grateful for the valuable comments from the anonymous reviewers on this manuscript.

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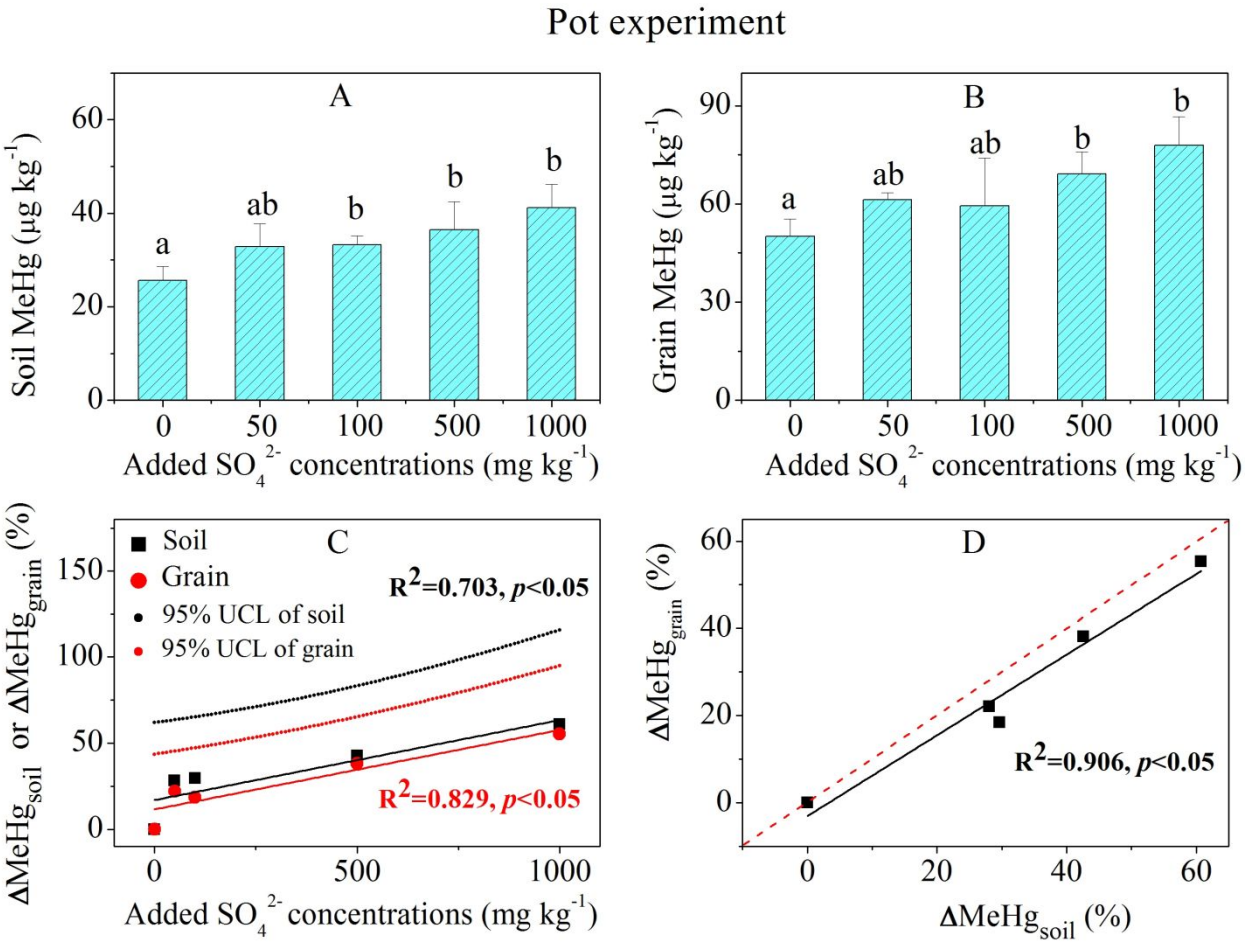
**Figure captions**

**Figure 1.** MeHg concentrations in soils (A) and rice grains (B) under sulfate amendment; the relationship between the changes in MeHg concentrations in grains ($\Delta MeHg_{grain}$) or soils ($\Delta MeHg_{soil}$) and added $SO_4^{2-}$ concentrations (C); the relationship between the changes in MeHg concentrations in grains ($\Delta MeHg_{grain}$) and soils ($\Delta MeHg_{soil}$) (D). Data shown are mean ± SD (n = 3). 95% UCL represents 95% upper confidence limit. Different letters above the bars indicate significant differences among treatments ($p$<0.05).

**Figure 2.** Changes in dissolved total Hg (THg) concentrations from HgS particles (A) and Hg-spiked Yixing (YX) soil (B), in response to increased ratios between amended sulfide and total Hg in solid. Data shown are mean ± SD (n = 3).

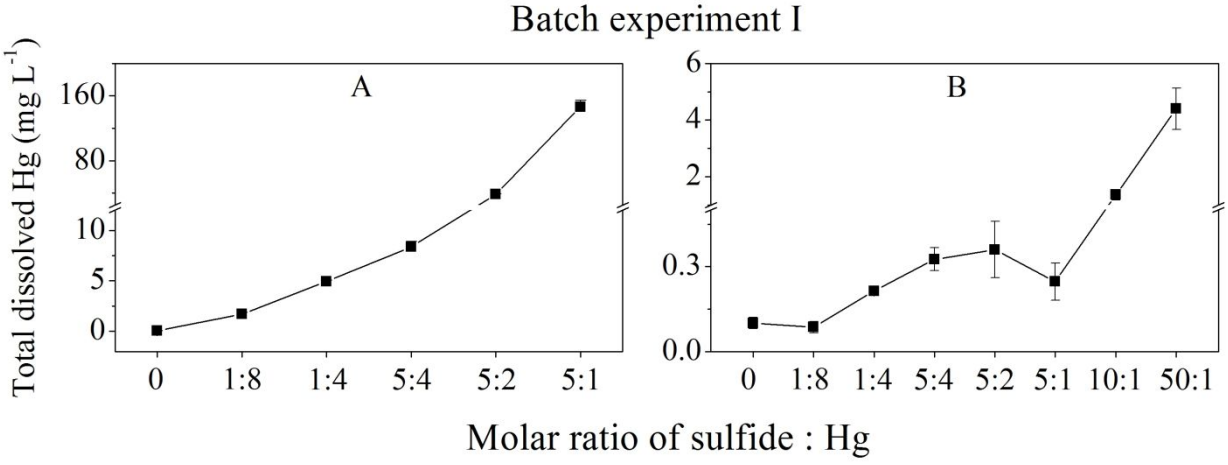
**Figure 3.** Dynamics of MeHg concentrations in soils during incubation. A, B, C represents WS, XY and QZ soil. Data shown are mean ± SD (n = 3). Different letters above the bars indicate significant differences among treatments ($p$<0.05).

730 **Figure 1**



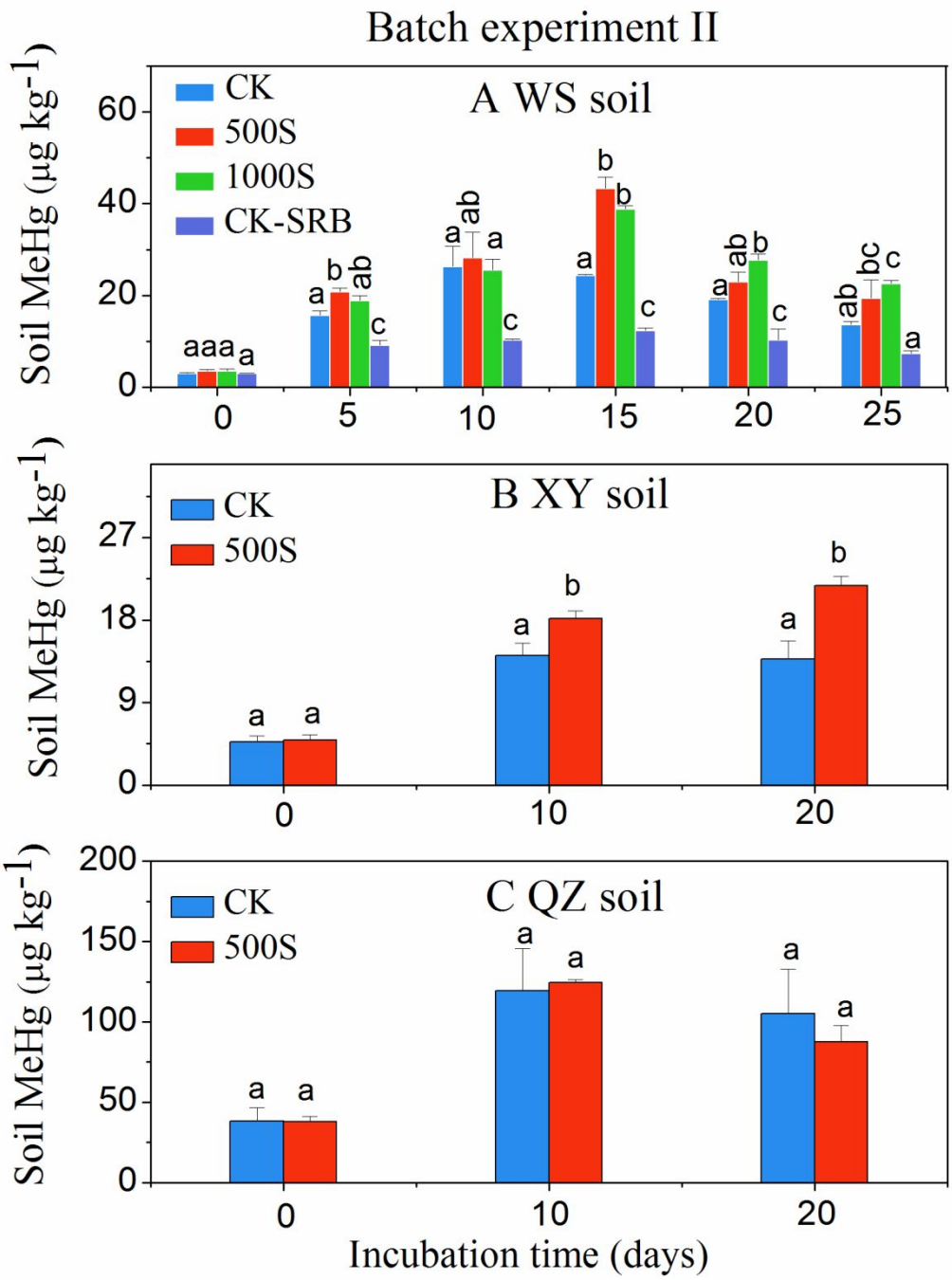

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733 **Figure 2**




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736 **Figure 3**




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